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# Primary and secondary stroke prophylaxis in children with sickle cell anemia: a meta-analysis

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We conducted a pooled analysis to evaluate the effectiveness of hydroxyurea (HU), chronic blood transfusion (CBT), myeloablative allogeneic hematopoietic stem cell transplant (HCT) with a matched related donor, and revascularization surgery (RVS) in children with sickle cell anemia (SCA). We compared the interventions with no therapy for primary and secondary stroke prevention. The Medline and EMBASE databases were searched for articles that included ≥6 patients published before January 2025. Four reviewers reviewed all studies to arrive at a consensus assessment. The stroke rates (per 100 person-year) for primary stroke prevention in children with transcranial Doppler measurement ≥200 cm/s were as follows: no therapy, 10.7; after initial HU, 1.0; and after initial CBT, 1.0. The stroke recurrence rates for secondary stroke prevention per 100 person-years were 19.6 after no therapy, 3.5 after initial HU, 2.7 after initial CBT, 1.0 after HCT, and 3.3 after RVS. Neither the HCT nor the RVS study adjusted for the time-dependent stroke recurrence rate, resulting in an overestimation of their therapeutic benefit compared to the stroke recurrence rates before the intervention. In low- and middle-income countries (LMICs), where ~99% of all children with SCA are born, the stroke recurrence rate in untreated children in the first year was 36.4 to 51.4 strokes per 100 person-years, in 2 studies; in the only randomized controlled trial for secondary stroke prevention with initial HU (SPRING trial n = 101), 38% (5 of 13) of the deaths occurred within the first year. For children in LMICs where local HU costs <$0.20 per day, HU is a practical, inexpensive option for primary and secondary stroke prevention.

## Introduction

The initial report of stroke in a child with sickle cell anemia (SCA) was described in 1923.[1] Over the last 4 decades, medical care for children with SCA has evolved from no treatment to evidence-based strategies for primary and secondary stroke prevention in both low- and middle-income countries (LMICs). In 2020, the American Society of Hematology panel published guidelines on preventing, diagnosing, and treating cerebrovascular disease in SCA.[2] As a focused effort, we performed a systematic review of primary and secondary overt stroke prevention strategies in children and adolescents (referred to as children). When appropriate, we stratified the studies into LMICs and high-income

Submitted 9 September 2025; accepted 24 September 2025; prepublished online on *Blood Advances* First Edition 10 October 2025; final version published online 2 February 2026. https://doi.org/10.1182/bloodadvances.2023012544.

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The full-text version of this article contains a data supplement.

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countries (HICs) because 99% of all children born live in LMICs (see supplemental Text 1 for the method used in calculations and the resulting spreadsheet). We included the following therapeutic interventions: chronic blood transfusion (CBT) therapy, hydroxyurea (HU), revascularization surgery (RVS), and myeloablative hematopoietic stem cell transplant (HCT) with a matched related donor.

## Methods

The systematic review was guided by the following population, intervention, comparison, and outcome questions.

1. In children and young adults with SCA without a prior stroke history and an abnormal transcranial Doppler (TCD) velocity ≥200 cm/s (nonimaging), how did the use of HU, CBT, or myeloablative matched related donor HCT, compared to no intervention, reduce the incidence rate of initial stroke?
2. In children and young adults with SCA with a prior history of stroke, how did the use of HU, CBT, myeloablative matched related donor HCT, or RVS, compared to no intervention, reduce the incidence rate of stroke recurrence?

Details of the search strategy, including a comprehensive list of search terms, meta-analysis/synthesis methods, study selection, and data extraction, are provided in supplemental Text 1. See Figure 1 for the study selection flow diagram.

## Results

### No therapy for primary stroke prevention

One study met our criteria for no treatment for primary stroke prevention based on abnormal TCD. In the Stroke Prevention for Sickle Cell Anemia (STOP) randomized controlled trial (RCT) by Adams et al,[3] there were 67 children and adolescents randomized to receive no therapy for primary stroke prevention (≥200 cm/s with nonimaging TCD), and the study was conducted in an HIC. Of these 67 participants, 10 strokes occurred. The pooled stroke rate for this study was 10.7 events per 100 person-years (95% confidence interval [CI], 5.4-19.2), with no deaths.[3]

### Initial HU therapy for primary stroke prevention

Five studies met our study criteria for the initial use of HU for primary stroke prevention based on abnormal TCD. This included 1 RCT,[4] 1 multicenter retrospective study,[5] 1 single-center retrospective study,[6] and 2 prospective single-center longitudinal studies.[7,8] Two studies were from HICs,[5,6] whereas the rest were from LMICs. Stroke was centrally adjudicated in 3 studies.[4,5,7] In total, the cohort included 337 children and adolescents for primary stroke prevention while receiving HU, with 12 strokes (11 overt and 1 transient ischemic attack [TIA]). The total person-years for the entire cohort were 1153.3. The pooled stroke rate was 1.0 event per 100 person-years (95% CI, 0.4-1.6; $P = .001$). There was no evidence of heterogeneity (Q = 1.74; df, 4; $P = .783$; $I^2 = 0.0$). There was no evidence of publication bias by Egger test ($P = .060$). Four studies reported death, with 15 deaths reported in Abdullahi et al,[4] 3 in Lagunju et al,[8] and none in Galadanci et al and Gulbis et al (see supplemental Table 1).[5,7] The pooled death rate was 1.1 per 100 person-years (95% CI, 0.1-2.1; $P = .031$).

### Initial CBT therapy for primary stroke prevention

Five studies met our inclusion criteria for the initial use of CBT for primary stroke prevention based on abnormal TCD. Our cohort included 2 RCTs. In the STOP RCT, participants were randomly assigned to receive chronic transfusions (n = 63) or the standard of care (n = 66).[3] In the TCD with Transfusions Changing to Hydroxyurea (TWiTCH) trial,[9] participants were randomly allocated to either chronic transfusions (n = 61) or HU (n = 60).

Three of 5 studies were single-center retrospective studies.[10-12] All were from an HIC. Central adjudication of neuroimaging was performed in the 2 RCTs.[3,9] In a total of 183 children across these studies, there were 4 strokes (1 overt and 3 TIAs). The total person-years for these 5 studies were 523.1. The pooled stroke

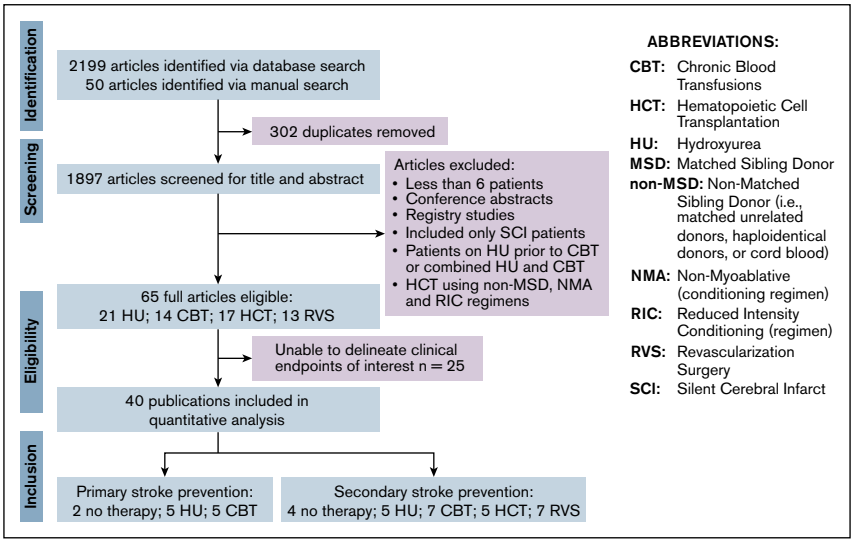


**Figure 1. Flow diagram for the systematic review of the literature to determine the efficacy of primary and secondary stroke prevention in children with SCA.**

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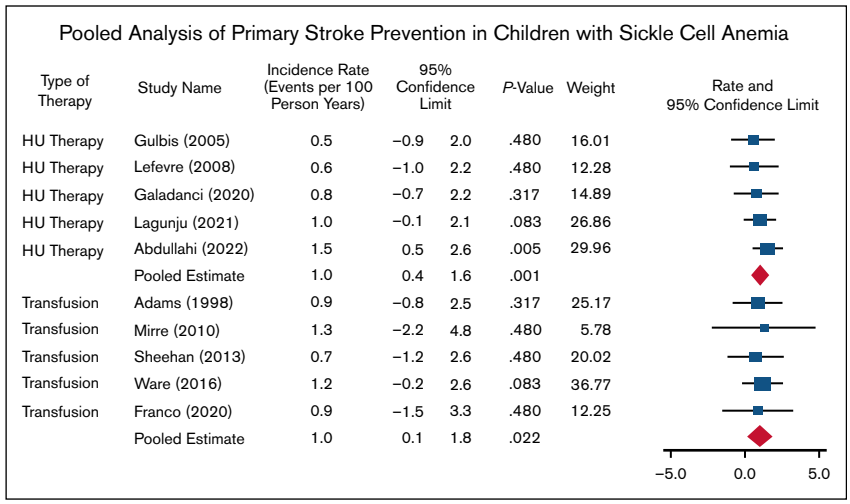
**Figure 2. Pooled analysis of primary stroke prevention in children with SCA.** Primary stroke prevention treatments include HU and transfusions. Each horizontal line represents the 95% CI for the incidence rate of stroke for those with TCD velocity ≥200 cm/sec in an individual study. The center of each box is the stroke event per 100 person-years. The size of the box is proportional to the study's weight in the meta-analysis. Diamonds indicate pooled estimates for each treatment category; the width of each diamond corresponds to the 95% CI, with a random-effects model to account for between-study heterogeneity.




rate was 1.0 event per 100 person-years (95% CI, 0.1-1.6; $P = .022$). There was no evidence of heterogeneity ($Q = 0.258$; *df*, 4; $P = .992$; $I^2 = 0$). There was no evidence of publication bias by Egger test ($P = .747$). Three studies reported no deaths,[3,9,11] and 2 studies did not comment on death.[10,12] The pooled death rate was 0.3 per 100 person-years (95% CI, −0.2 to −0.8; $P = .304$).

## HCT for primary stroke prevention

Four studies used HCT for primary stroke prevention.[13-16] We were unable to evaluate the studies because we could not identify the subgroup of children with abnormal TCD velocities as the primary indication for HCT and their subsequent neurological outcomes.

## Elevated TCD velocity can be a risk factor for stroke with or without treatment

In addition to the 10 ischemic stroke events in the STOP trial by Adams et al,[3] 6 additional patients developed a stroke during extended follow-up, as reported by Lee et al[17] (data not included in the analysis). All 6 patients had a baseline TCD ≥220 cm/s; the last interpretable TCD was abnormal, regardless of their transfusion status. Five of these patients were from the no-therapy group, and 1 was from the transfusion group. The 1 patient from the transfusion group who had developed a stroke discontinued transfusions after the trial period. Her TCD velocity was normal during transfusions but reverted to abnormal 6 months after stopping transfusions. We were unable to include these patients from the extended follow-up because the no-therapy group started transfusion therapy, and the 1 person who had a stroke from the transfusion group stopped transfusions after the initial STOP Trial.

In the Stroke Prevention in Nigeria study, 8 children had strokes after initiating HU.[4] A higher proportion of strokes was observed in those who had a TCD velocity ≥220 cm/s (4/8 [50%]) than in those with a TCD velocity <220 cm/s (3/205 [1%]; $P < .0001$). Similarly, in Lagunju et al, 2 strokes occurred in children in whom abnormal TCD velocities persisted even after HU treatment.[8] Both children were receiving HU therapy for >28 months, and both had a TCD velocity ≥220 cm/s at the last interpretable TCD before stroke. Both studies show that there is a subset of children whose TCD velocities remain abnormal despite receiving HU therapy, confirming that this group has a significant stroke risk. Thus, persistently elevated TCD velocity despite treatment with CBT or HU is a risk factor for stroke, even while receiving therapy. Figure 2 shows the pooled analysis of primary stroke prevention strategies in children with SCA.

## No therapy for secondary stroke prevention

Four studies met the inclusion criteria for no therapy for secondary stroke prevention, 1 prospective single-center longitudinal study[18] and 3 retrospective single-center studies.[19-21] Three of these studies were from LMICs,[18-20] and 1 was from an HIC.[21] None of these studies performed any prospective neuroimaging or central adjudication of strokes.

The study by Powars et al included 1 patient who received intermittent transfusions for secondary stroke prevention, a therapeutic approach considered as the current standard care; however, this patient could not undergo effective transfusion and was counted as such by the authors.[21] Based on available data, Powars et al reported that the stroke recurrence rate for ≤12 months and ≤24 months was 14.2 and 15.2 strokes per 100 person-years, respectively (see supplemental Table 2).[21]

Two studies from LMICs reported stroke recurrence rates in the no-therapy group. Lagunju et al reported that 9 of 18 patients (50%) had a recurrent stroke ≤12 months of the initial stroke, corresponding to a stroke recurrence rate of 51.4 strokes per 100 person-years.[19] Ali et al reported a stroke recurrence rate of ~36.4 strokes per 100 person-years ≤12 months after the initial stroke.[18] The estimate in the Ali et al study is imprecise because the results provided a percentage of strokes after >12 months (59%), not the actual number of patients.[18] Using the percentage to calculate the number of patients, we found that there were 11.8, which we rounded to 12 for analysis.

Without any treatment, in the Powars et al, Ali et al, and Lagunju et al studies, ≥46% of children had a recurrent stroke within 36 months of the initial stroke: 46% (7/15), 88% (~18/20), and 78% (14/18), respectively.[18,19,21] The study by Cunningham et al

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did not report the timing of stroke recurrence.[20] A total of 98 children and adolescents had an initial stroke, received no therapy, and were observed for a total of 404.6 person-years. A total of 73 overt strokes (no TIAs) occurred, with a pooled stroke recurrence rate of 19.6 per 100 person-years (95% CI, 8.9-30.4; $P < .001$). There was evidence of heterogeneity (Q = 18.9; *df*, 3; $P < .001$), related to the significantly lower recurrent stroke rate in Powars et al ($I^2 = 84.1\%$).[21] The pooled analysis showed no evidence of publication bias by Egger test ($P = .069$). Among 98 untreated children with strokes, 4 patients died in the Ali et al study, 2 in the Lagunju et al study, and 6 in the Cunningham-Myrie et al study, for a total of 12 deaths, resulting in a pooled death rate of 2.8 per 100 person-years (95% CI, 0.0-5.7; $P = .054$).[18-20]

## Initial HU therapy for secondary stroke prevention

Five studies met the inclusion criteria for the initial use of HU for secondary stroke prevention: 1 randomized controlled study,[22] 1 prospective single-center longitudinal study,[18] 1 multicenter retrospective registry-based study,[5] and 2 retrospective single-center studies.[19,20] Only 1 study was from an HIC.[4] The remaining studies were from LMICs.[17-19,21] The dose of HU ranged from a low dose of 10 mg/kg per day to the maximum tolerated dose. Prospective neuroimaging was not performed in any of these studies, and stroke was centrally adjudicated only in 2 studies.[5,22]

The timing of stroke recurrence in those treated with initial HU therapy indicated that the initial 12- and 24-month period after the stroke carried the highest risk of stroke recurrence. Ali et al did not report any stroke ≤12 months of the initial stroke but reported that 1 of 10 patients (10%) had a stroke recurrence ≤24 months of the initial stroke, corresponding to a stroke recurrence rate of 5.2 strokes per 100 person-years.[18] Lagunju et al reported no recurrent overt stroke ≤12 months of the initial overt stroke but reported that 2 of 13 patients (15.4%) had a stroke recurrence ≤24 months of the initial stroke, corresponding to a stroke recurrence rate of 8.2 strokes per 100 person-years.[19] Abdullahi et al reported a stroke recurrence rate of 8.5 and 6.3 strokes per 100 years for ≤12 and ≤24 months after the initial stroke, respectively.[22] We could not determine the timing of recurrent stroke in 1 study.[20] One study examined initial HU in an HIC and showed no stroke recurrence ≤24 months.[5]

A total of 142 children initially treated with HU for an overt stroke were included in the cohort. The total person-years were 500.7 among the 5 studies. Among this cohort, 19 children had a second overt stroke (no TIAs). The pooled stroke recurrence rate was 3.5 events per 100 person-years (95% CI, 1.9-5.1; $P < .001$). There was no evidence of heterogeneity (Q = 1.45; *df*, 4; $P = .836$; $I^2 = 0$). There was no evidence for publication bias by Egger test ($P = .828$). There were 13 deaths across these 5 studies, all in a prospective RCT,[22] resulting in a pooled mortality rate of 2.3 per 100 person-years (95% CI, 1.0-3.7; $P = .001$).

## Initial CBTs for secondary stroke prevention

Seven studies from HICs met the inclusion criteria for initial CBTs for secondary stroke prophylaxis. This included 1 prospective RCT (SWiTCH [Stroke with Transfusions changing to HU] trial),[23] 1 prospective multicenter study,[24] 1 prospective single-center study,[25] 1 retrospective multicenter study,[26] and 3 single-center retrospective studies.[11,27,28] In the SWiTCH trial, patients randomized to CBT had no strokes (n = 66) compared to a 10% stroke incidence in patients randomly allocated to the maximum tolerated dose of HU (these patients received CBT and chelation before randomization; n = 67)[23]; we included only the randomly allocated participants who started and completed CBT. Cohen et al reported an ischemic stroke a few months after stopping transfusions and 2 hemorrhagic strokes, which we included.[25] Ware et al and Hubert et al performed central adjudication of neuroimages and neurological events.[23,24] The remaining 5 studies did not have central adjudication of neuroimages and neurological events. A total of 357 children were observed for a cumulative 2420.0 person-years.

The timing of stroke recurrence for those treated with initial blood transfusion therapy indicated that the initial 12- and 24-month period after the stroke carried the highest risk of stroke recurrence. Hulbert et al reported that 6 of 40 patients (15%) had a recurrent stroke (4 overt and 2 TIA) ≤12 months of the initial stroke, corresponding to a stroke recurrence rate of 16.0 strokes per 100 person-years[24]; and 8 of 40 (20%) had a stroke recurrence ≤24 months of the initial stroke, corresponding to a stroke recurrence rate of 11.2 strokes per 100 person-years. Scothorn et al reported a recurrent stroke rate of 4.0 strokes per 100 person-years in ≤24 months; however, this study excluded children who did not receive at least 5 years of regular blood transfusion therapy, with a median and mean follow-up time of ~10 years.[28] Majumdar et al reported a stroke recurrence rate of 0 and 6.3 strokes per 100 person-years for ≤12 and ≤24 months after the initial stroke, respectively.[27] Cohen et al and Mirre et al reported no stroke recurrences in ≤24 months but did report an ischemic stroke a few months after stopping transfusions and 2 hemorrhagic strokes, which we included.[11,25] We were unable to determine the timing of stroke recurrence in Ware et al and Pegelow et al.[23,26]

Among a total cohort of 357 children with an initial stroke, 65 children had recurrent strokes (53 overt and 12 TIAs) while receiving CBT, resulting in a pooled stroke rate of 2.7 per 100 person-years (95% CI, 1.4-4.0; $P < .001$). There was evidence of heterogeneity (Q = 18.9; *df*, 6; $P = .004$; $I^2 = 68.3\%$). There was no evidence of publication bias by Egger test ($P = .124$). A total of 15 deaths occurred in the studies, including 6 hemorrhagic strokes, with a pooled mortality rate of 0.6 per 100 person-years (95% CI, 0.3-0.9; $P < .001$).

## HCT for secondary stroke prophylaxis

Five studies met the inclusion criteria for HCT for secondary stroke prevention. These included 2 single-center retrospective studies,[16,29] 1 prospective single-center study,[15] and 2 prospective multicenter studies.[13,30] All studies were from HICs. Prospective neuroimaging was performed only in 3 studies after transplant.[13,15,30] Central adjudication of neuroimaging was only reported by Carpenter et al and in 7 of 29 patients by Walter et al.[16,30] A total of 91 patients had HCT for secondary stroke prevention, observed for a total of 565.8 person-years. Of these 91 children, 4 had overt strokes and 1 TIA. Among the 4 strokes, 2 were hemorrhagic, and 2 were ischemic. The pooled stroke rate was 1.0 strokes per 100 person-years (95% CI, 0.2-1.8; $P = .020$). There was no evidence of heterogeneity (Q = 0.58; *df*, 4; $P = .966$; $I^2 = 0.0\%$). There was no evidence of publication bias by Egger test ($P = .190$). There was a total of 5

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deaths, with a pooled mortality rate of 0.6 per 100 person-years (95% CI, 0.0-1.2; $P = .065$).

## RVS

Seven studies met the inclusion criteria for RVS as a secondary stroke prevention measure. We excluded children with SCA who underwent RVS for the prevention of headaches or for abnormal TCD velocities. There were 2 prospective[31,32] and 5 retrospective studies.[33-37] All studies were from HICs. Only 1 study performed prospective neuroimaging and central adjudication of strokes after RVS.[32]

Five different revascularization procedures were used in this cohort; however, we were unable to analyze patients according to procedures due to the small numbers in each RVS procedure group. The most common revascularization procedure was pial synangiosis (n = 47), followed by encephalo-duro-arterio-synangiosis (n = 36). Other approaches included encephalo-myo-arterio-synangiosis (n = 6), dural inversion (n = 2), and encephalo-duro-arterio-myosynangiosis (n = 1). A total of 57 children had a first-time revascularization procedure for secondary stroke, observed for a total of 269.5 person-years. Among these 57 children, there were 12 strokes (3 overt and 9 TIAs). Of these 12 strokes, 8 (67%; 2 overt and 6 TIAs) occurred within the first month after surgery, and 4 (33%; 1 overt and 3 TIAs) happened beyond the first month after RVS. The pooled stroke recurrence rate after RVS was 3.3 strokes per 100 person-years (95% CI, 1.2-5.5; $P = .003$). There was no evidence of heterogeneity (Q = 3.48; *df*, 6; $P = .746$; $I^2 = 0.0\%$). There was no evidence of publication bias by Egger test ($P = .076$).

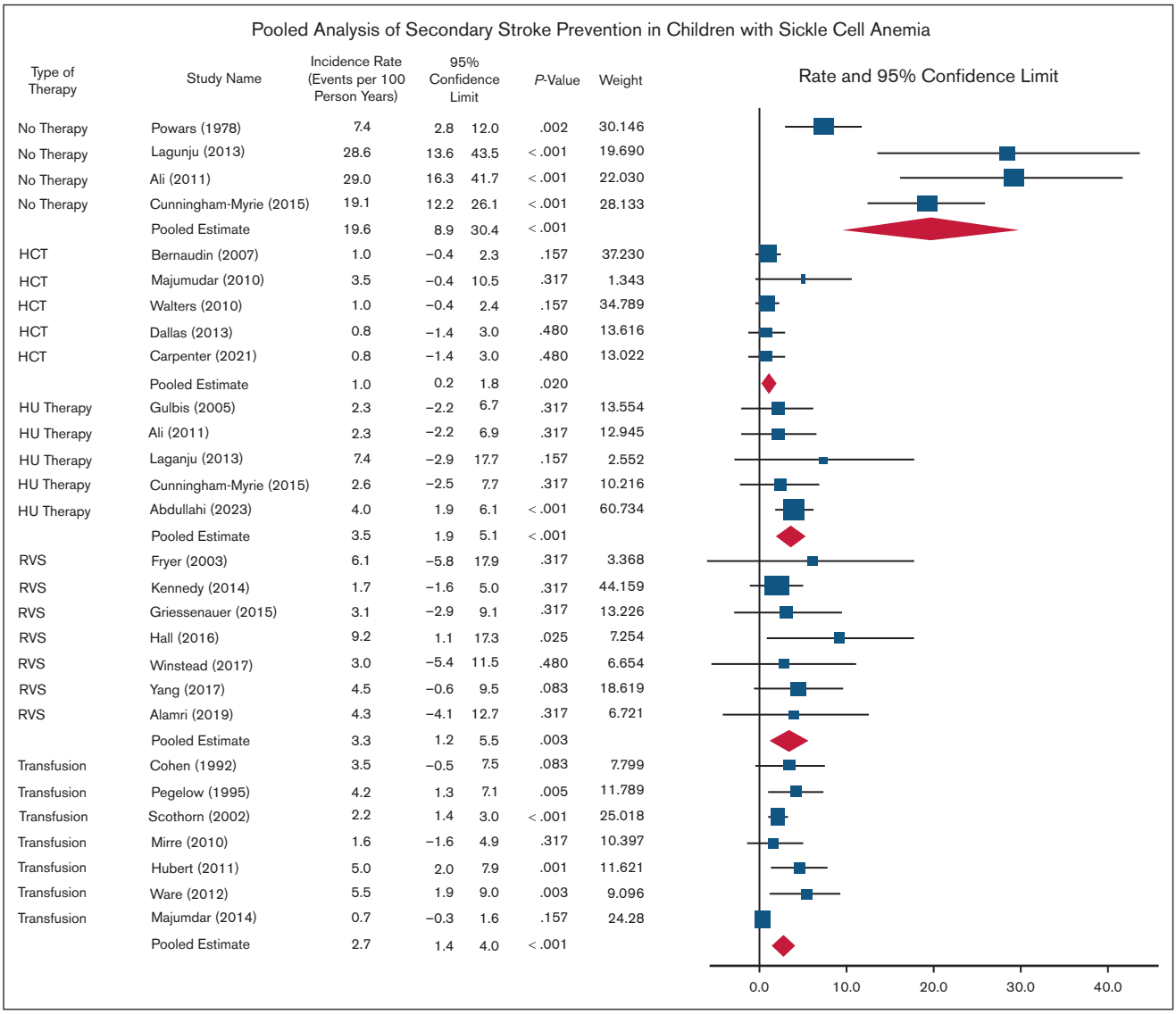


**Figure 3. Pooled analysis of secondary stroke prevention in children with SCA, with the reference population of children with no therapy at the top.** Each horizontal line represents the 95% CI for the incidence rate of recurrent stroke in an individual study. The center of each box is the stroke recurrence event per 100 person-years. The size of the box is proportional to the study's weight in the meta-analysis. Diamonds indicate pooled estimates for each treatment category; the width of each diamond corresponds to the 95% CI, with a random-effects model to account for between-study heterogeneity.

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A total of 2 deaths were reported after RVS, resulting in a pooled mortality rate of 1.4 per 100 person-years (95% CI, 0.0-2.7; $P = .055$). Figure 3 has the pooled analysis of secondary stroke prevention in children with SCA.

## Discussion

Overt stroke remains one of the most debilitating and life-threatening neurological complications in children with SCA. RCTs provide evidence-based strategies for children living in LMICs and HICs.[2] In our systematic review, for primary and secondary stroke prevention in children living in LMICs, initial HU treatment, as low as 10 mg/kg per day, is an effective strategy compared to no therapy. For children living in HICs, the optimal management for primary and secondary stroke prevention is initial blood transfusion therapy to maintain a hemoglobin level >9.0 g/dL and a hemoglobin S level <30%. If CBT is not available as the initial strategy for primary and secondary stroke prevention, HU is a reasonable alternative compared to no therapy. Furthermore, the pooled analysis revealed that for primary stroke prevention, initial HU yields a stroke recurrence incidence rate similar to that of initial regular CBT therapy.

The pooled analysis revealed 2 novel findings that will immediately affect the medical care of children with SCA in LMICs. First, the stroke recurrence rate in the first year was the highest in untreated children living in LMICs, ranging from 36.4 to 51.4 strokes per 100 person-years in Ali et al[18] and Lagunju et al.[19] Second, we noted a high proportion of children died in the only randomized trial (the SPRINT trial (n = 101) of secondary stroke prevention using initial HU.[22] In this trial, 5 of 13 deaths (38%) occurred within the first year, indicating the importance of close initial patient surveillance after a stroke, potentially even monthly visits for the first 12 to 24 months after a stroke. For children living in LMICs who remain untreated for secondary stroke prevention, we have a unique opportunity to prevent lifelong morbidity and mortality with low-dose HU (10 mg/kg per day) for <$0.20 per day.[38]

Our pooled analysis also revealed another novel observation that should at least be considered in patient management: persistently abnormal TCD values ≥200 cm/s (nonimaging; CareFusion Sonara/tek, San Diego, CA), regardless of treatment (HU or CBT), are associated with an increased rate of stroke. Until recently, standard care has not included TCD velocity measurement after starting CBT or HU; however, RCTs in Nigeria and the United States demonstrated that persistently elevated TCD velocities after starting HU or CBT were associated with an ongoing risk of an overt stroke. Based on these data, a practical approach would be to perform TCD measurements 3 months after initiating therapy with HU or CBT for all children with abnormal TCD velocities. If the TCD velocity remains elevated, reevaluate adherence and consider increasing the dose of HU or CBT. One potential option to lower persistent TCD velocities after therapy is with HCT. Bernaudin et al demonstrated that HCT is an effective strategy for decreasing persistently elevated TCD velocities after treatment with HCT.[39]

Our study has several limitations. We could not compare the incidence rates of stroke recurrence for HCT or RVS with those for regular blood transfusion (a perceived standard of care in HICs for secondary stroke prevention). We demonstrated that, in children with strokes, the incidence of recurrent strokes is the highest in the first 24 months after the stroke. Most RVS or HCT studies were conducted beyond the high-risk period of stroke recurrence. To improve statistical inferences when comparing CBT to RVS or HCT for secondary stroke prevention, future studies must address the time-dependent stroke recurrence rate in children. Without acknowledging that the stroke recurrence rate is time dependent, an estimate of the stroke recurrence rate for RVS or HCT compared to CBT will likely overestimate the benefit of RVS or HCT, as was the case in this pooled analysis.

Another unavoidable challenge for curative therapy is that we only included outcomes after myeloablative matched-sibling donor transplant because the American Society of Hematology panel on guidelines for transplant in sickle cell disease suggested consideration of an HLA-matched related transplant for secondary stroke prevention.[40] Given these limitations, we recommend that primary and secondary stroke prevention studies adjudicate neurological events and report their neurological outcomes similarly to randomized controlled stroke prevention trials in children with SCA.[3,23,41] We did not include children with sickle cell disease who received gene therapy in our analysis, because most gene therapy clinical trials have excluded patients with a history of cerebrovascular disease.

In conclusion, our pooled data analysis has an immediate impact on the care of children with SCA living in LMICs, demonstrating that untreated strokes and the first 12 to 24 months after a stroke are critical to prevent stroke recurrence and death, respectively. We also confirmed the conclusions of the American Society of Hematology 2020 guidelines for primary and secondary stroke prevention.

## Acknowledgments

This study was supported by funding from National Institute of Neurological Disorders and Stroke grant 5R01NS094041-02 and National Heart, Lung, and Blood Institute grant 1U01HL156620-01.

## Authorship

Contribution: M.R.D. conceived the idea of the manuscript; H.G.R. and J.A.M.-F. wrote the initial draft; and all authors reviewed the various versions of the drafts and approved the final manuscript.

Conflict-of-interest disclosure: The authors declare no competing financial interests.

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